Retail Energy Operations Analytics

Billing Exceptions & Service Order Performance

Power BI | Power Query | DAX

Portfolio Case Study | 2026

Where are operational workload and performance issues concentrated?

Retail energy operations teams manage billing exceptions and field service orders across multiple processes and stakeholders. The challenge is identifying where workload, financial impact, ageing backlog and service-order performance issues are concentrated.

- Billing exception workload & financial impact
- Service-order outcomes & SLA performance
- Ageing backlog & overdue service orders

1,573
Exception Volume

Billing Exceptions

Exception Type	Exception Volume	Total Realised Financial Impact
Meter Reads	564	$3,103.54
Plans	345	$1,190.41
Payments	307	$999.1

1,174
Service Order Volume

19.34%
Unsuccesful SO Rate

Service Order Performance by Type

SO Type	Service Order Volume	Unsuccesful SO Rate
Meter Exchange	340	15.88%
Supply/Energisation	262	21.37%
Meter Configuration/Alteration	248	21.37%
Read Investigation	248	18.95%

How was the data structured for analysis?

Dataset

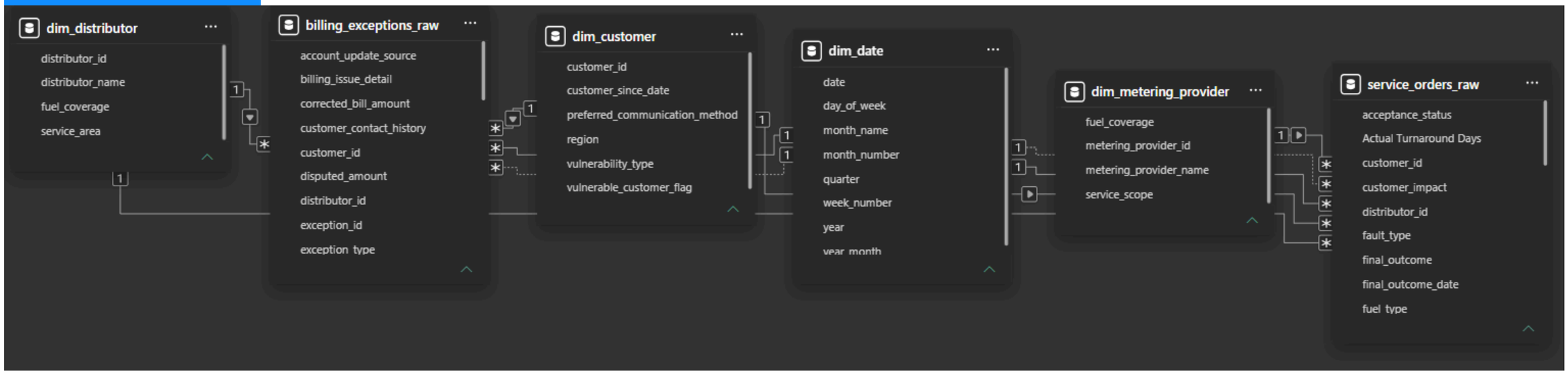
- Synthetic six-month retail energy dataset
- 1,573 billing exceptions
- 1,174 service orders

Preparation

- Power Query cleaning and transformation
- Standardised operational fields/categories
- Prepared fact and dimension tables

Model

- Star-schema structure
- Shared customer, date, distributor and metering-provider dimensions
- Shared date dimension with active/inactive relationships
- Supports raised/reported vs resolved/finalised analysis

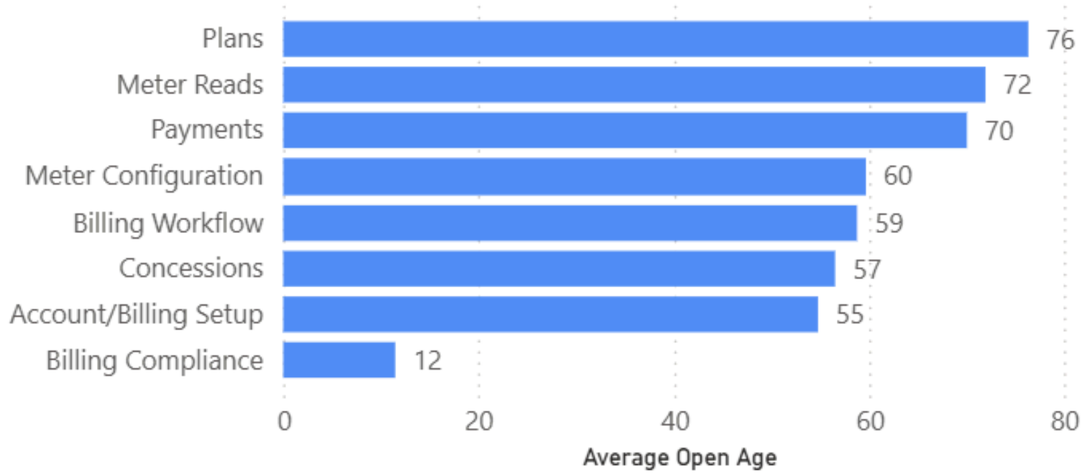


Meter Reads drive the largest billing workload and financial impact

Billing Exceptions by Type

Exception Type	Exception Volume	Total Realised Financial Impact
Meter Reads	564	$3,103.54
Plans	345	$1,190.41
Payments	307	$999.1
Meter Configuration	141	$267.44
Concessions	115	$534.54
Billing Workflow	47	$220.91
Account/Billing Setup	40	$66.95
Billing Compliance	14	$215.55
Total	**1,573**	**$6,598.44**

Average Open Age by Type



564
Meter Read exceptions

76 days
Average open age for Plans

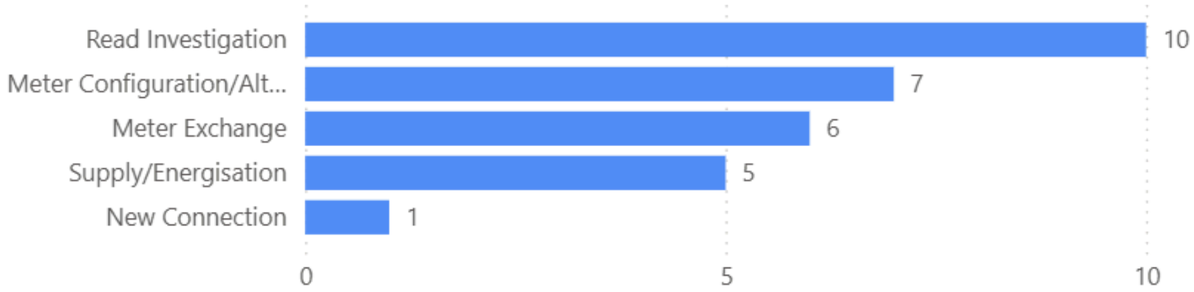
Meter Reads lead both workload and realised financial impact, while **Plan-related exceptions** show the oldest unresolved backlog.

Supply/Energisation leads SLA breaches; Read Investigations dominate overdue work

Service Order Performance by Type

SO Type	Service Order Volume	Unsuccesful SO Rate	SLA Breach Rate
Meter Exchange	340	15.88%	13.35%
Supply/Energisation	262	21.37%	15.08%
Meter Configuration/Alteration	248	21.37%	8.47%
Read Investigation	248	18.95%	11.64%
New Connection	49	26.53%	11.63%
Supply Abolishment	27	14.81%	11.11%
Total	**1,174**	**19.34%**	**12.23%**

Overdue Open SOs by Type



15.08%
Supply/Energisation SLA breach rate

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Overdue open SOs are Read Investigations

Supply/Energisation combines elevated unsuccessful outcomes with the highest SLA breach rate, while **Read Investigations** account for the largest overdue backlog.

Four operational priorities emerge from the analysis

01 — Reduce Meter Read exceptions
Investigate repeated estimates and missing-read drivers with metering and distributor stakeholders.

02 — Clear ageing Plan/Rate backlog
Prioritise aged Plan/Rate cases and review retailer-side workflow and data hand-offs contributing to repeat exceptions.

03 — Improve service-order readiness
Strengthen job-information validation, customer contact and site-access checks before execution.

04 — Focus backlog and SLA performance
Prioritise overdue Read Investigations and investigate Supply/Energisation execution and SLA outcomes.

Recommendations prioritise **workload reduction**, **backlog clearance** and **preventable service-order failures**.